

1. Bank Account Management System:

Problem Statement: Design a system to manage bank accounts.

Classes: Account, Savings-Account, Current-Account.

Attributes: Account number, balance, account type, interest rate (for savings account), overdraft limit (for current account), etc.

Methods: Deposit, Withdraw, Calculate Interest, Transfer, etc.

Object Usage: Create objects for each customer's account and perform transactions like deposits, withdrawals, and transfers.

Steps:

Start the program.

Create a base class Account with account number, balance, and account type.

Add methods for deposit, withdrawal, transfer, and displaying account details.

Create SavingsAccount by extending Account.

Add interest calculation to the savings account.

Create CurrentAccount by extending Account.

Override the withdraw() method to support overdraft limit.

Create Savings and Current account objects.

Display the bank menu.

Read the user's choice.

Perform the selected operation using switch-case.

Repeat the menu until the user selects Exit.

Close the Scanner.

Stop the program.

Algorithm:

Step 1: Start.

Step 2: Create Account class with account number, balance, and type.

Step 3: Define deposit(), withdraw(), transfer() and show() methods.

Step 4: Create SavingsAccount extending Account and define calcInterest().

Step 5: Create CurrentAccount extending Account and override withdraw().

Step 6: Create Savings and Current account objects.

Step 7: Display the menu and read the choice.

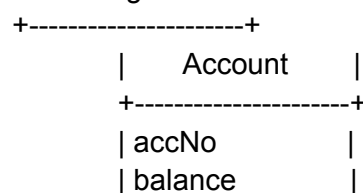
Step 8: Perform the selected operation using switch-case.

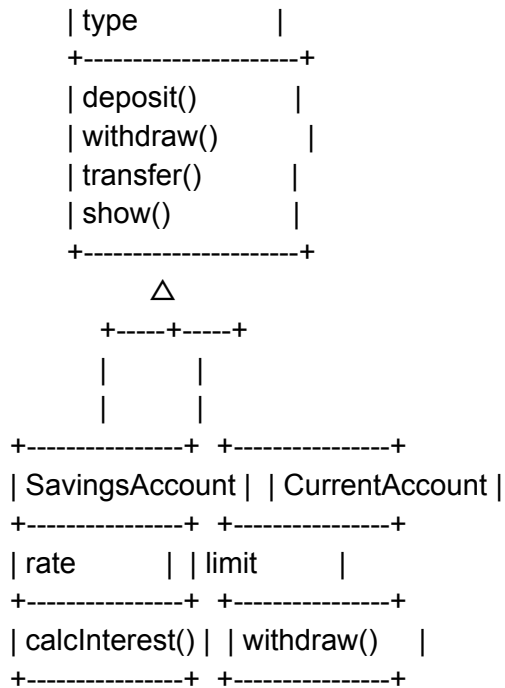
Step 9: Repeat until the user chooses 9. Exit.

Step 10: Close Scanner.

Step 11: Stop.

UML Diagram:





Code:

```
package javacore;
```

```
import java.util.Scanner;
```

```
// Base class for all accounts
```

```
class Account {
```

```
    int accNo;
```

```
    double balance;
```

```
    String type;
```

```
    // Constructor
```

```
    Account(int no, double bal, String t) {
```

```
        accNo = no;
```

```
        balance = bal;
```

```
        type = t;
```

```
    }
```

```
    // Deposit money
```

```
    void deposit(double amt) {
```

```
        balance += amt;
```

```
        System.out.println("Deposited: " + amt);
```

```
        System.out.println("Balance: " + balance);
```

```
    }
```

```
// Withdraw money
void withdraw(double amt) {
    if (amt <= balance) {
        balance -= amt;
        System.out.println("Withdrawn: " + amt);
        System.out.println("Balance: " + balance);
    } else {
        System.out.println("Not enough balance!");
    }
}
}
```

```
// Transfer money to another account
void transfer(Account other, double amt) {
    if (amt <= balance) {
        balance -= amt;
        other.balance += amt;
        System.out.println("Transfer done!");
    } else {
        System.out.println("Transfer failed!");
    }
}
}
```

```
// Show account information
void show() {
    System.out.println("Acc No : " + accNo);
    System.out.println("Type : " + type);
    System.out.println("Bal : " + balance);
}
}
```

```
// Savings account class
class SavingsAccount extends Account {

    double rate;

    SavingsAccount(int no, double bal, double r) {
        super(no, bal, "Savings");
        rate = r;
    }

    // Calculate interest
    void calcInterest() {
        double interest = balance * rate / 100;
    }
}
```

```
        System.out.println("Interest: " + interest);
    }
}
```

// Current account class

```
class CurrentAccount extends Account {
```

```
    double limit;
```

```
    CurrentAccount(int no, double bal, double l) {
        super(no, bal, "Current");
        limit = l;
    }
}
```

// Overriding withdraw

```
@Override
```

```
void withdraw(double amt) {
    if (amt <= balance + limit) {
        balance -= amt;
        System.out.println("Withdraw ok. Balance: " + balance);
    } else {
        System.out.println("Overdraft limit crossed!");
    }
}
}
```

// Main class

```
public class BankSystem {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

// Create objects

```
SavingsAccount sAcc =
    new SavingsAccount(101, 5000, 5);
```

```
CurrentAccount cAcc =
    new CurrentAccount(201, 3000, 2000);
```

```
int ch;
```

```
do {
    System.out.println("\n===== BANK ACCOUNT SYSTEM =====");
    System.out.println("1. Show Savings Account");
    System.out.println("2. Deposit in Savings");
    System.out.println("3. Withdraw from Savings");
    System.out.println("4. Calculate Interest");
    System.out.println("5. Show Current Account");
    System.out.println("6. Deposit in Current");
    System.out.println("7. Withdraw from Current");
    System.out.println("8. Transfer Money");
    System.out.println("9. Exit");

    System.out.print("Enter choice: ");
    ch = sc.nextInt();

    switch (ch) {

        case 1:
            sAcc.show();
            break;

        case 2:
            System.out.print("Enter amount: ");
            double d1 = sc.nextDouble();
            sAcc.deposit(d1);
            break;

        case 3:
            System.out.print("Enter amount: ");
            double w1 = sc.nextDouble();
            sAcc.withdraw(w1);
            break;

        case 4:
            sAcc.calcInterest();
            break;

        case 5:
            cAcc.show();
            break;

        case 6:
            System.out.print("Enter amount: ");
            double d2 = sc.nextDouble();
```

```
        cAcc.deposit(d2);
        break;

    case 7:
        System.out.print("Enter amount: ");
        double w2 = sc.nextDouble();
        cAcc.withdraw(w2);
        break;

    case 8:
        System.out.print("Enter amount to transfer: ");
        double t = sc.nextDouble();
        sAcc.transfer(cAcc, t);
        break;

    case 9:
        System.out.println("Thank you!");
        break;

    default:
        System.out.println("Invalid choice!");
    }

    } while (ch != 9);

    sc.close();
}
}
```

Output:

===== BANK ACCOUNT SYSTEM =====

1. Show Savings Account
2. Deposit in Savings
3. Withdraw from Savings
4. Calculate Interest
5. Show Current Account
6. Deposit in Current
7. Withdraw from Current
8. Transfer Money
9. Exit

Enter choice: 7

Enter amount: 40000

Overdraft limit crossed!

===== BANK ACCOUNT SYSTEM =====

1. Show Savings Account
2. Deposit in Savings
3. Withdraw from Savings
4. Calculate Interest
5. Show Current Account
6. Deposit in Current
7. Withdraw from Current
8. Transfer Money
9. Exit

Enter choice: 9

Thank you!